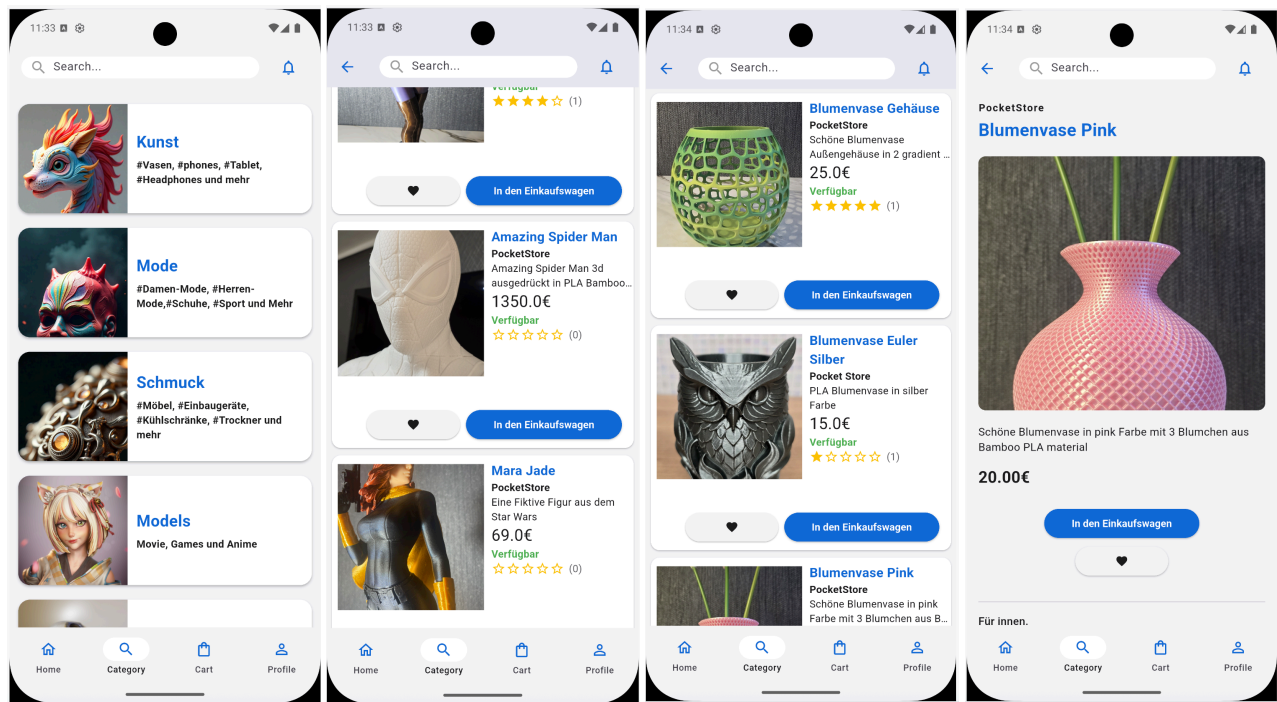


PocketStore

A modern Flutter-based e-commerce application specializing in 3D printed products with Firebase backend integration.

Application Screenshots

Image Library



Home Screen

Modern home interface with featured products and intuitive navigation

Product List View

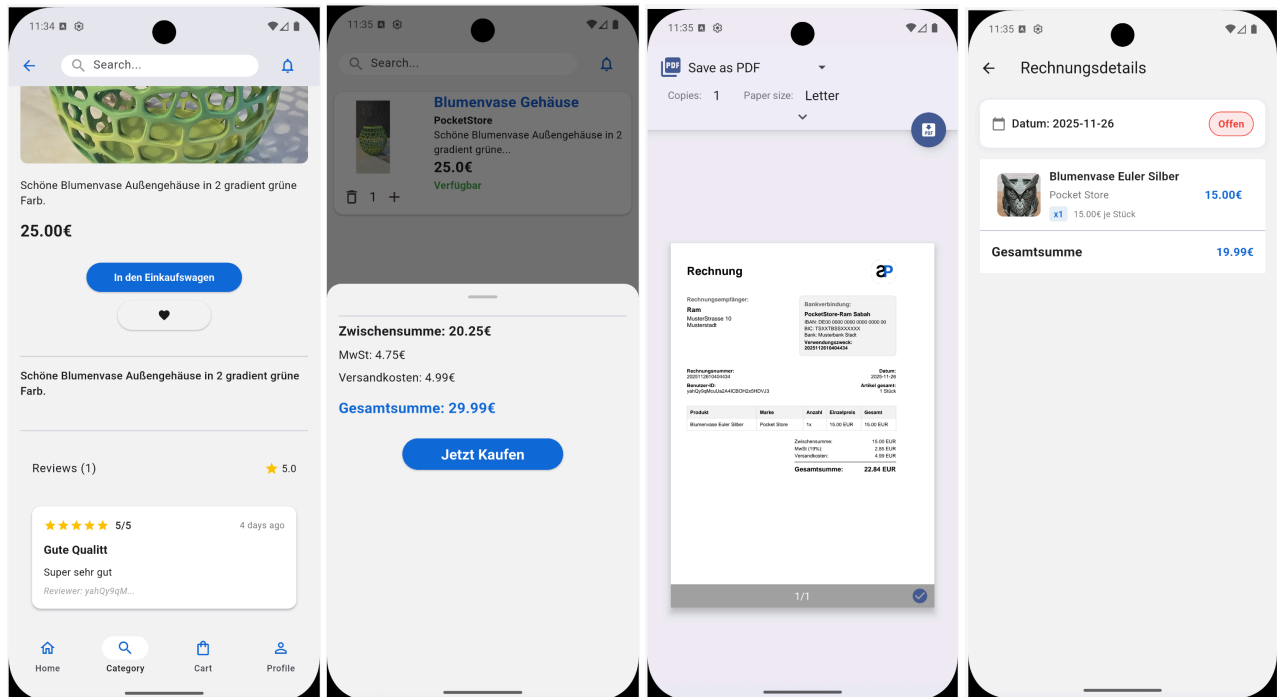
Product catalog showing items in card format

Category View

Product view in cards for selected category

Product Details

Comprehensive product information and specifications



Single Product View

Detailed product view with specifications and pricing

Shopping Cart

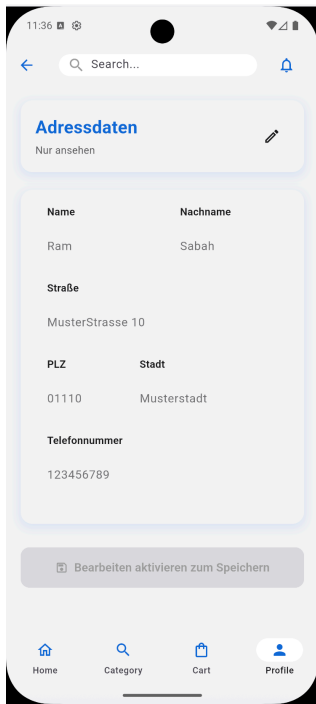
Smart cart system with quantity controls and pricing

Invoice Overview

Bill overview of ordered products with details

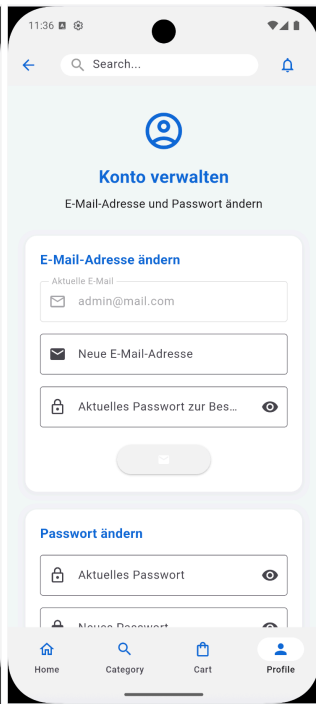
Order Overview

Overview of orders showing payment status



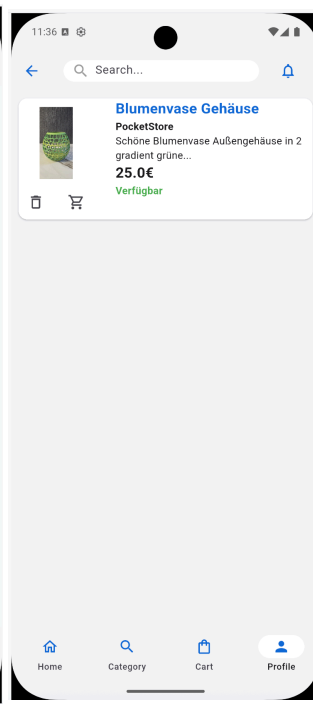
User Address Data

User address information and delivery details



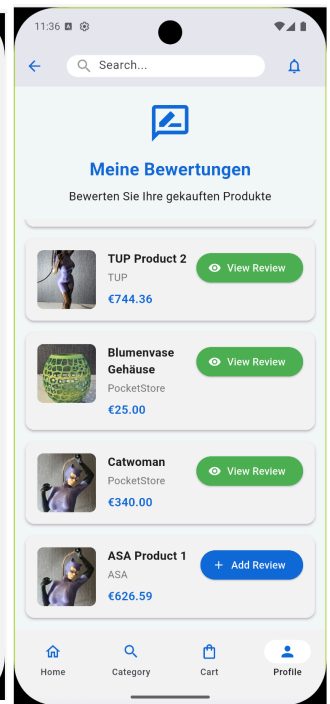
Account Settings

Personal information and preferences management



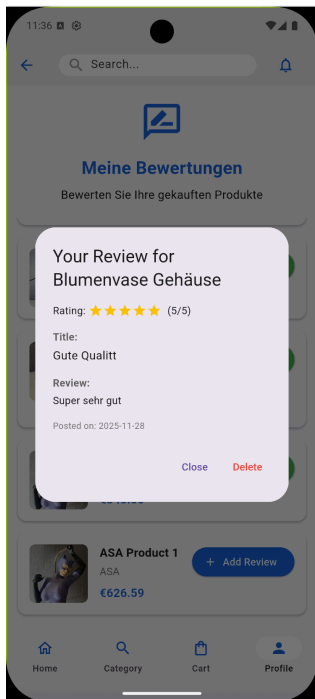
Favorites Overview

Overview of favorite products and wishlist



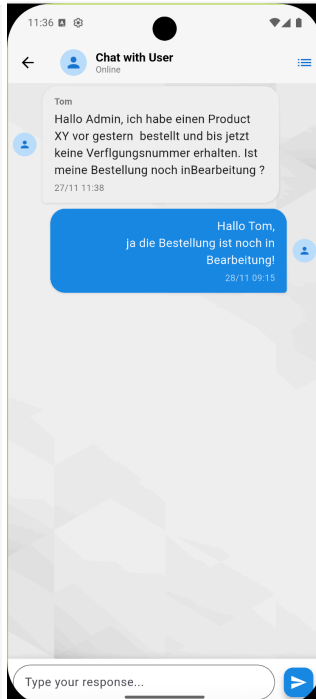
Product Reviews Overview

Overview of rated and reviewed products



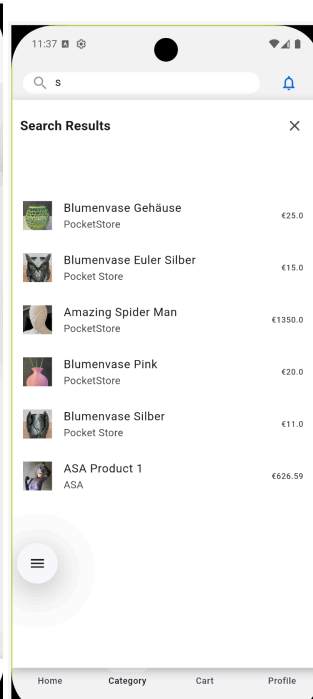
Written Reviews Overview

Overview of reviews written by the user



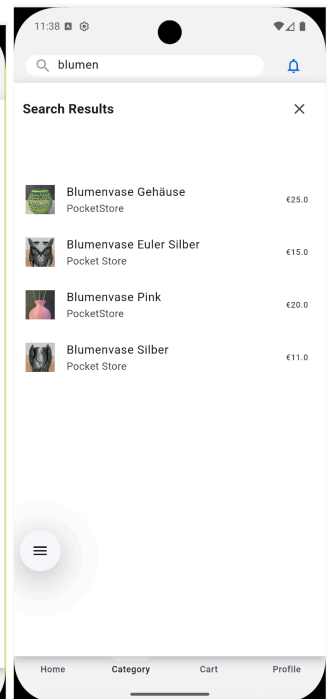
Customer-Admin Chat

Chat interface between customer and admin



Search Overview

Search interface with results display



Search Results View

Search overview with different search terms

🚀 Key Features

📱 Core Functionality

- **Modern Home Interface:** Clean and intuitive design with easy navigation
- **Product Catalog:** Browse products in card format by categories
- **Advanced Search:** Search products by name, brand, and category with live results
- **Product Details:** Comprehensive single product view with specifications

Shopping & Order Management

- **Smart Cart System:** Add products to cart with quantity management
- **Invoice Management:** Detailed bill overview of ordered products
- **Order Overview:** Track orders and view payment status
- **Order History:** Complete purchase tracking and status updates

User Management & Profile

- **Address Management:** Manage user delivery addresses and contact information
- **Account Settings:** Personal information and preferences management
- **Favorites System:** Save favorite products and manage wishlists
- **Review System:** Write and manage product reviews and ratings

Communication & Reviews

- **Customer Support Chat:** Real-time chat between customers and admin
- **Review Management:** Overview of written reviews and product ratings
- **Search Interface:** Advanced search functionality with multiple search terms
- **User Feedback:** Comprehensive review and rating system

Technical Stack

Frontend

- **Flutter:** Cross-platform mobile development
- **Dart:** Programming language
- **Material Design:** Modern UI components
- **State Management:** Stateful widgets and providers

Backend & Services

- **Firebase Authentication:** Secure user authentication
- **Cloud Firestore:** Real-time NoSQL database
- **Firebase Functions:** Serverless backend functions
- **Firebase Storage:** File and image storage

Key Dependencies

```
dependencies:  
  flutter: sdk: flutter  
  firebase_core: ^4.2.0  
  firebase_auth: ^6.1.1  
  cloud_firestore: ^6.0.3  
  cloud_functions: ^6.0.3  
  image_picker: ^1.2.0  
  uuid: ^4.5.1  
  in_app_purchase: ^3.2.1  
  printing: ^5.13.4  
  webview_flutter: ^4.10.1
```

Supported Platforms

-  **Android** (API 21+)
-  **iOS** (iOS 15.0+)
-  **Web** (Progressive Web App)
-  **macOS** (Desktop support)

Getting Started

Prerequisites

- Flutter SDK (3.0.0 or higher)
- Dart SDK (3.0.0 or higher)
- Firebase project setup
- Android Studio / Xcode for device testing

Installation

1. Clone the repository

```
git clone https://github.com/ramsabah28/application2.git
cd application2
```

2. Install dependencies

```
flutter pub get
```

3. Configure Firebase

- Add your google-services.json (Android) to android/app/
- Add your GoogleService-Info.plist (iOS) to ios/Runner/
- Update lib/firebase_options.dart with your Firebase configuration

4. Run the application

```
flutter run
```



Project Structure

```
lib/
├── main.dart           # Application entry point
├── firebase_options.dart # Firebase configuration
├── src/
│   ├── component/      # UI components
│   │   ├── features/   # Feature-specific widgets
│   │   ├── profile/    # Profile-related screens
│   │   └── homeScreenFeatures/ # Home screen components
│   ├── data/           # Data layer and utilities
│   ├── models/         # Data models
│   ├── repository/     # Data repositories
│   ├── services/       # Business logic services
│   └── payment/        # Payment processing
```



Key Features Implementation

3D Model Integration

- Interactive 3D product visualization
- Multiple viewing angles and zoom capabilities
- Cross-platform 3D rendering support

Real-time Features

- Live search with instant results
- Real-time stock updates
- Live chat support
- Push notifications for order updates

E-commerce Functionality

- Secure payment processing
- Inventory management
- Order tracking system
- Multi-currency support
- Tax calculation

User Experience

- Offline capability
- Dark/Light theme support

- Multi-language support (German interface)
- Accessibility features
- Responsive design

Security Features

- **Firebase Authentication:** Multi-factor authentication support
- **Secure Payments:** PCI DSS compliant payment processing
- **Data Encryption:** End-to-end encryption for sensitive data
- **Input Validation:** Comprehensive input sanitization
- **Session Management:** Secure session handling

Performance Optimizations

- **Image Optimization:** Compressed images with lazy loading
- **Caching Strategy:** Smart caching for better performance
- **Database Optimization:** Efficient Firestore queries
- **Code Splitting:** Modular code architecture
- **State Management:** Optimized state updates

Testing

```
# Run unit tests
flutter test

# Run integration tests
flutter test integration_test/

# Generate coverage report
flutter test --coverage
```

Future Enhancements

- ☐ AI-powered product recommendations
- ☐ Augmented Reality (AR) product preview
- ☐ Voice search functionality
- ☐ Advanced analytics dashboard
- ☐ Multi-vendor marketplace support
- ☐ Subscription-based services

Contributing

1. Fork the repository
2. Create a feature branch (`git checkout -b feature/AmazingFeature`)
3. Commit your changes (`git commit -m 'Add some AmazingFeature'`)
4. Push to the branch (`git push origin feature/AmazingFeature`)
5. Open a Pull Request

License

This project is licensed under the MIT License - see the [LICENSE](#) file for details.

Support

For support and questions: -  Email: support@pocketstore.com -  Live Chat: Available in the app -  Documentation: [Wiki](#)

Changelog

Version 1.0.0

- Initial release with core e-commerce functionality
- 3D model integration
- Firebase backend integration
- Cross-platform support

Made with ❤️ using Flutter and Firebase